



Mini Research Project

Sub series matters

Collatz Conjecture

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Supervised by Jun.-Prof. Hofmann

TABLE OF CONTENTS

01 Background

02 Previous work

03 Our Contribution

04 Scaling using parallel computing

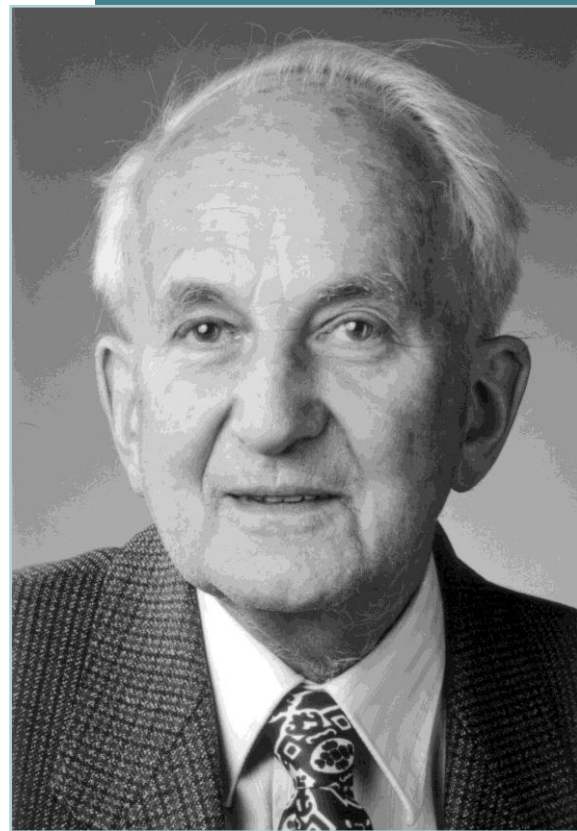


Unsolved mathematical problem

Collatz Conjecture (1937)

For every positive integer n , repeated application of the map $T: \{1, 2, 3, \dots\} \rightarrow T: \{1, 2, 3, \dots\}$ defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod{2}, \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2}, \end{cases}$$



Lothar Collatz 1910 – 1990

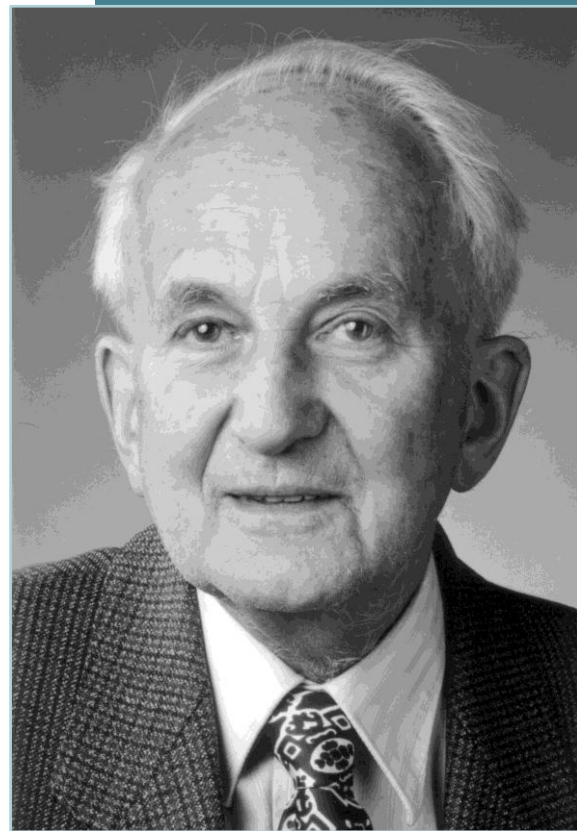
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repeated application of $T(n)$ eventually reaches the value 1



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Unsolved mathematical problem

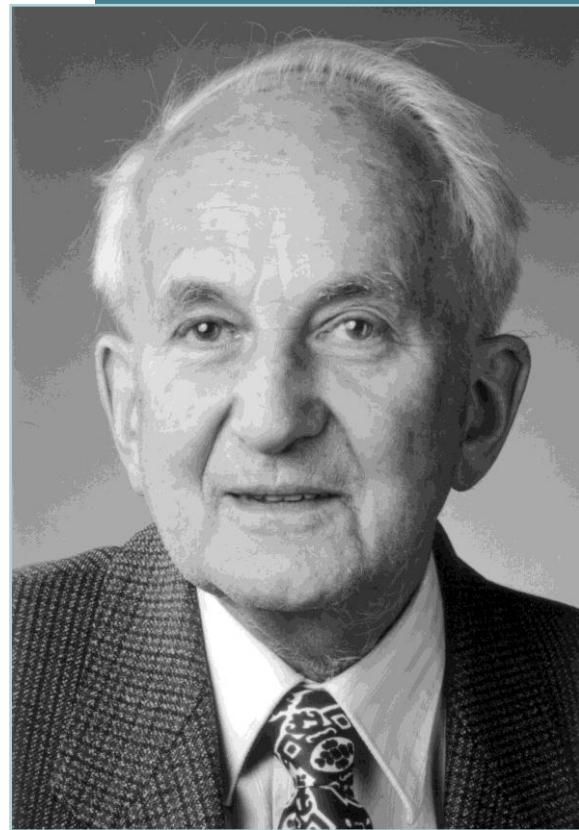
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Examples : $5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

$20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$



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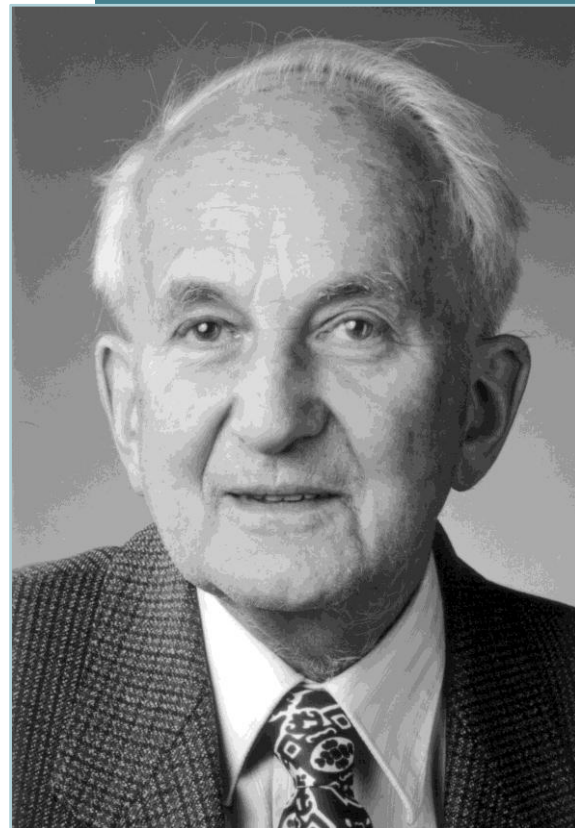
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Examples : $5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

$20 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$

$\vdots$

n Is this holds for all n ?



Lothar Collatz 1910 – 1990

02

Contributions

Mathematical Approaches

Proof for almost all positive integers

Terras (1976), Acta Arithmetica



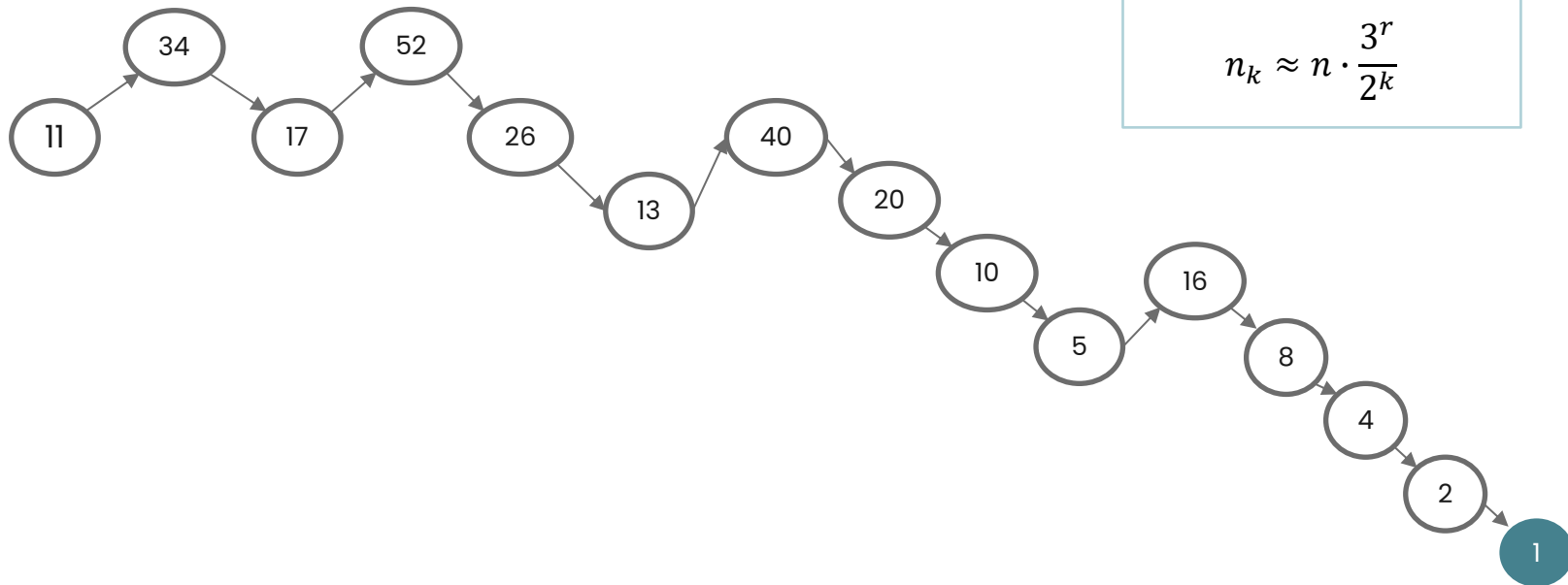
Mathematical Approaches

Proof for almost all positive integers

Terras (1976), Acta Arithmetica

Terra's analysis

$$n_k \approx n \cdot \frac{3^r}{2^k}$$



Mathematical Approaches

Proof for almost all positive integers

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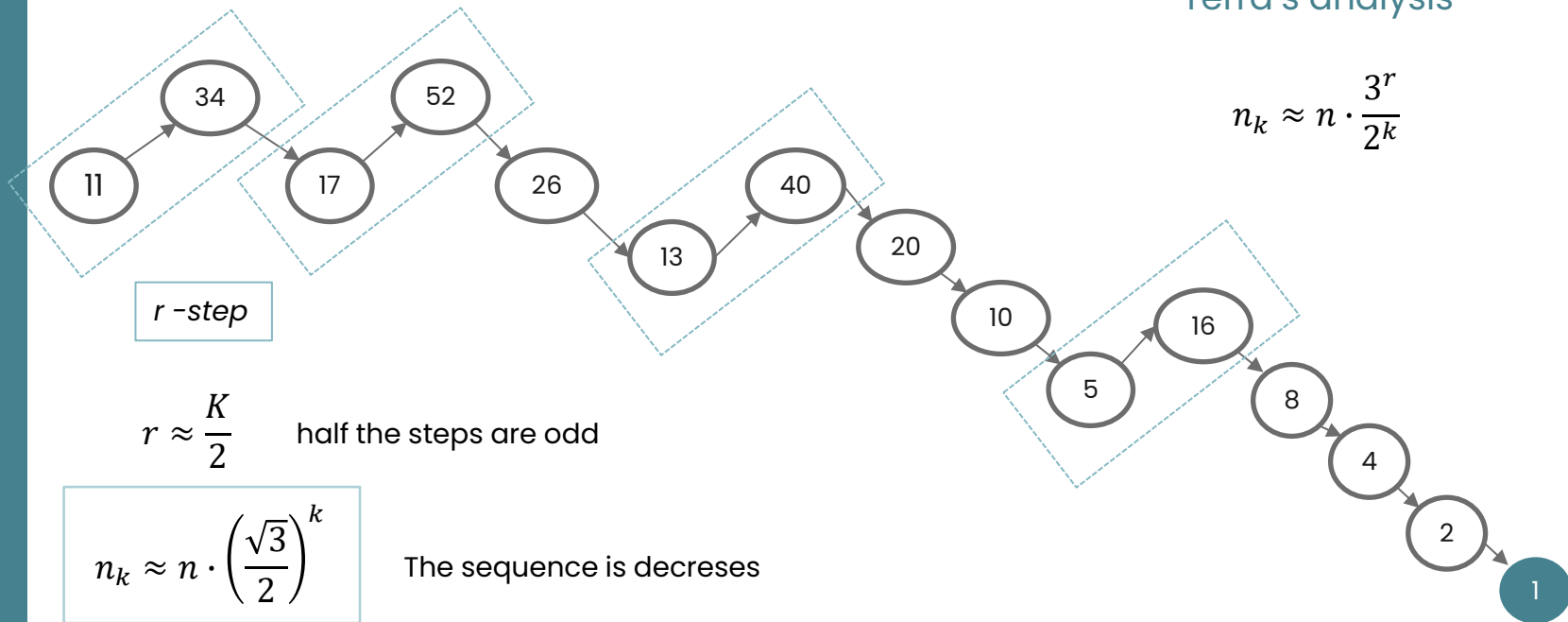
$$n_k \approx n \cdot \frac{3^r}{2^k}$$

r -step

$r \approx \frac{K}{2}$ half the steps are odd

$$n_k \approx n \cdot \left(\frac{\sqrt{3}}{2}\right)^k$$

The sequence is decreases



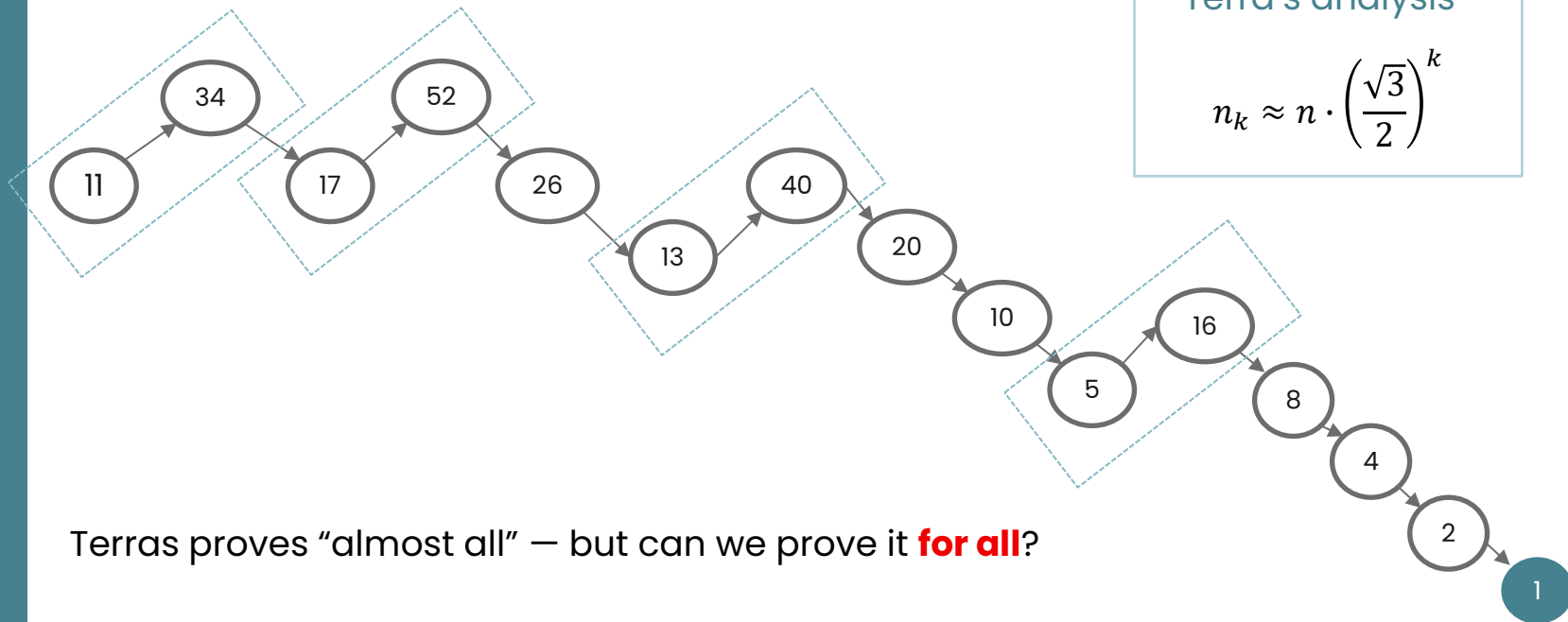
Mathematical Approaches

Proof for almost all positive integers

Terras (1976), Acta Arithmetica

Terra's analysis

$$n_k \approx n \cdot \left(\frac{\sqrt{3}}{2}\right)^k$$



Terras proves “almost all” — but can we prove it **for all**?

~~Proofs~~ Claims

*Diedrich (2025) – A Complete
Proof of the Collatz
Conjecture*

7. Conclusion

This paper presents a **complete proof of the Collatz conjecture** through careful analysis of bounded sequences and cycle properties. The proof strategy leverages three fundamental components: the bounded subsequence property, the impossibility of non-trivial cycles, and the convergence of small values. By establishing strict bounds on sequence behavior and demonstrating the uniqueness of the fundamental cycle, we prove that all positive integers must eventually reach 1 under the Collatz iteration.

*Izadi (2021) – Complete
Proof of Collatz's
Conjectures*

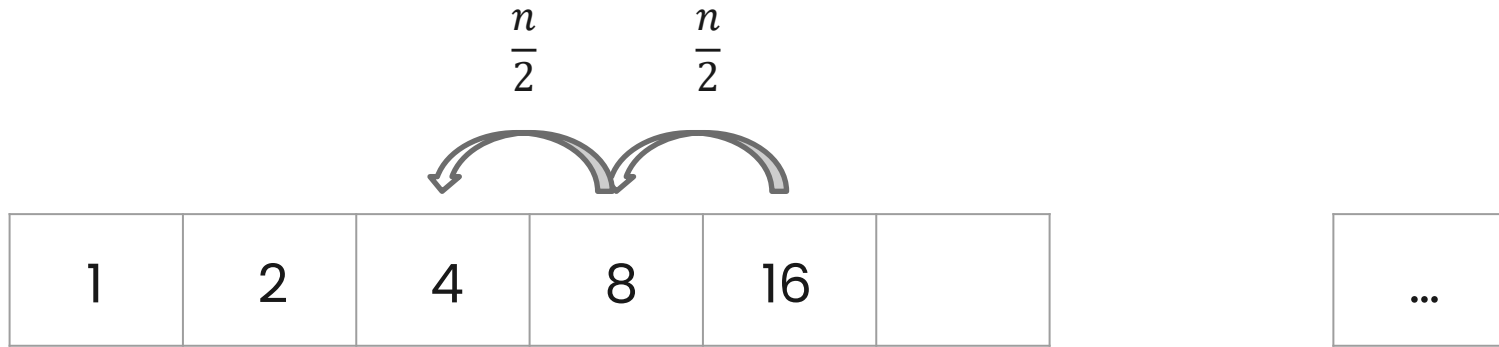
Then $f(k) = 6(27m + 3i) + 5 = 6h + 5$, where $h = 27m + 3i$. BY the part III of the first proof, it follows that $f(k) \in E$ and consequently $k \in E$. We clearly see that the density of the numbers in this proof also equals 1, and **the proof is complete**.

*Opfer (2011) – An Analytic
Approach to the Collatz $3n + 1$
Problem*

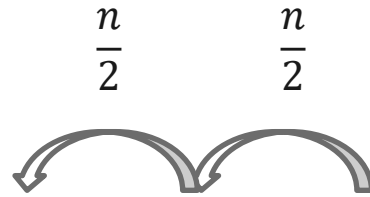
□

If we have a look at Theorem 2.4, and compare it with Conclusion 4.18, we see that **the Collatz conjecture is true**. This result is essentially based on the analytic approach of the Collatz $3n + 1$ problem by Berg and Meinardus, [1, 2].

Computational Approaches



Computational Approaches



1	2	4	8	16	
3	6	12	24	48	
5	10	20	40	80	
7	14	28	56	112	

...
...
...
...

Computational Approaches

m

$m \cdot 2^i$

1	2	4	8	16	
3	6	12	24	48	
5	10	20	40	80	
7	14	28	56	112	

...
...
...
...

1	2	4	8	16	
---	---	---	---	----	--

$$3 \times 1 + 1$$

$$3 \times 5 + 1$$

3	6	12	24	48	
---	---	----	----	----	--

5	10	20	40	80	
---	----	----	----	----	--

$$3 \times 3 + 1$$

$$3 \times 13 + 1$$

7	14	28	56	112	
---	----	----	----	-----	--

$$3 \times 9 + 1$$

$$3 \times 37 + 1$$

1	2	4	8	16	
---	---	---	---	----	--

1		5
---	--	---

3	6	12	24	48	
---	---	----	----	----	--

$$3(2n + 1)$$

5	10	20	40	80	
---	----	----	----	----	--

3		13
---	--	----

7	14	28	56	112	
---	----	----	----	-----	--

9		37
---	--	----

1	2	4	8	16	
---	---	---	---	----	--



3	6	12	24	48	
---	---	----	----	----	--

$$3(2n + 1)$$

5	10	20	40	80	
---	----	----	----	----	--



7	14	28	56	112	
---	----	----	----	-----	--



1	2	4	8	16	
---	---	---	---	----	--



3	6	12	24	48	
---	---	----	----	----	--

$$3(2n + 1)$$

5	10	20	40	80	
---	----	----	----	----	--



7	14	28	56	112	
---	----	----	----	-----	--



Computational Approaches

m →

$\frac{m \cdot 2^i - 1}{3}$

5	7	11	13	17	19	23	25	29	31	35	37
3	9	7	17	11	25	15	33	19	41	23	49
13	37	29	69	45	101	61	133	77	165	93	197
53	149	117	277	181	405	245	533	309	661	373	789
213	597	469	1109	725	1621	981	2133	1237	2645	1493	3157
853	2389	1877	4437	2901	6485	3925	8533	4949	10581	5973	12629
3413	9557	7509	17749	11605	25941	15701	34133	19797	42325	23893	50517

Computational Approaches

The conjecture has been verified **computationally**
for all $n < 2^{71}$

*Barina (2025) – Improved Verification Limit for the Convergence
of the Collatz Conjecture, Journal of Supercomputing*

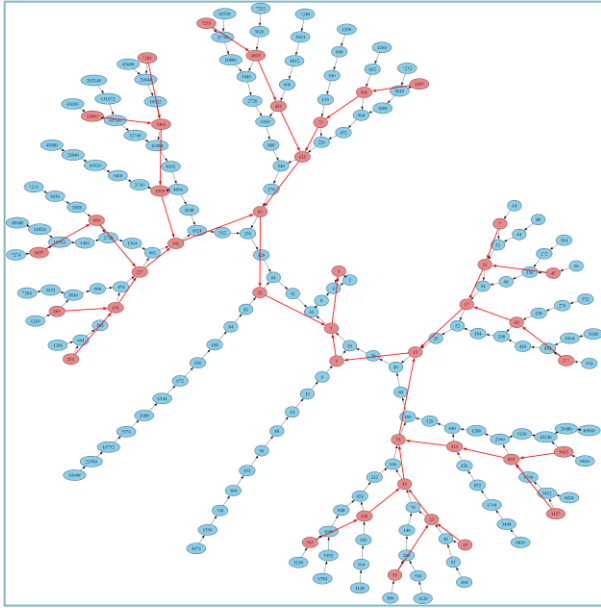
$$2^{71}$$

The conjecture has been verified The
computationally for individual numbers as large as

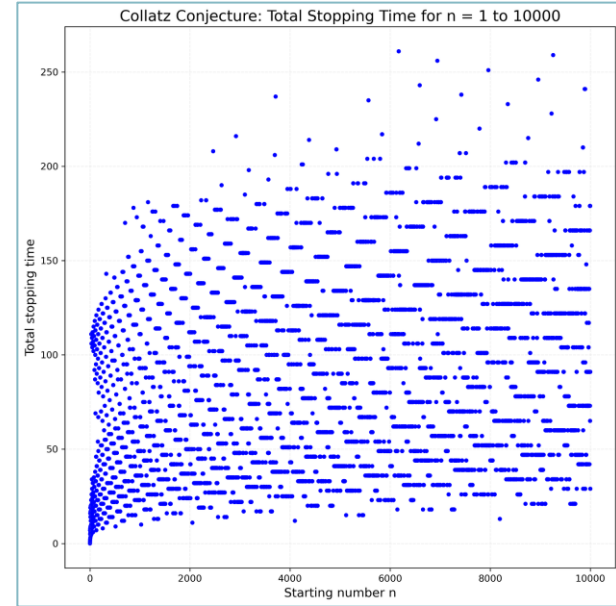
Ren et al. (2018) – *Collatz conjecture for $2^{100000} - 1$ is true: algorithms for verifying
extremely large numbers*. IEEE SmartWorld

$$2^{100000}$$

Chaos of Collatz sequence



Collatz Tree



Collatz stopping time

Order in Chaos

The sub-series $\{3(2n + 1)\}$ do appear in later lines

5	7	11	13	17	19	23	25	29	31	35	37
3	9	7	17	11	25	15	33	19	41	23	49
13	37	29	69	45	101	61	133	77	165	93	197
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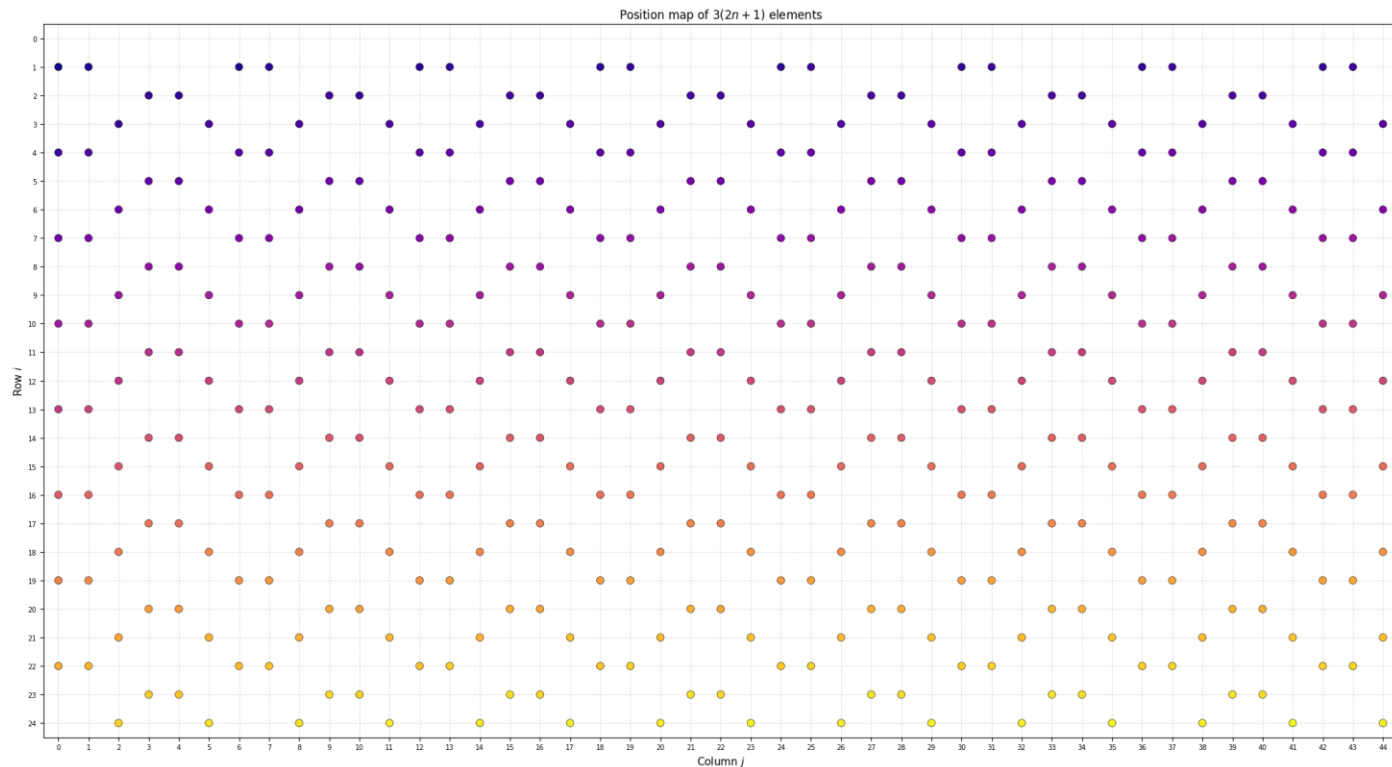
Order in Chaos

The sub-series $\{3(2n + 1)\}$ appear display a periodic structure

5	7	11	13	17	19	23	25	29	31	35	37
3	9	7	17	11	25	15	33	19	41	23	49
13	37	29	69	45	101	61	133	77	165	93	197
53	149	117	277	181	405	245	533	309	661	373	789
213	537	469	1109	725	1621	981	2133	1237	2645	1493	3157
853	2389	1877	4437	2951	6485	3925	8533	4949	10581	5973	12629
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Order in Chaos

the **wave pattern** of the sub-series $3(2n + 1)$



Order in Chaos

What is the maximum we reached ?

Max → 9
For 5×5
matrix

3
45
69
213
597

5	7	11	13	17	19	23	25	29	31	35	37
3	9	7	17	11	25	15	33	19	41	23	49
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Order in Chaos

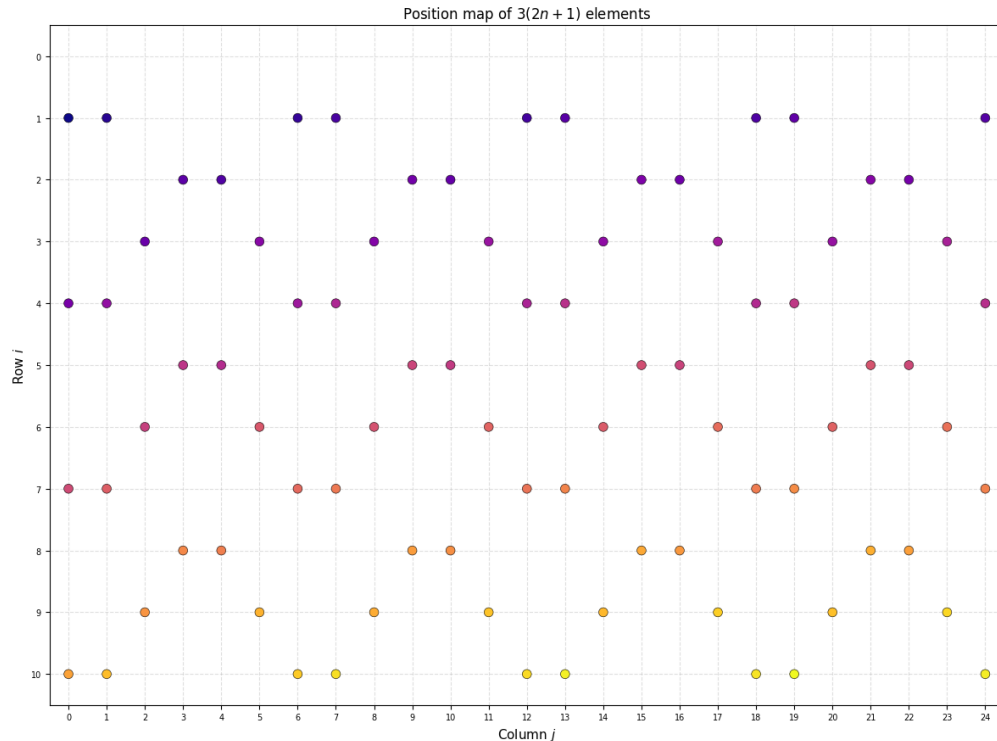
What is the maximum we reached ?

Max
For 6×8
matrix

3
9
15
21
33
45
69
213
597
981
2133

5	7	11	13	17	19	23	25	29	31	35	37
3	9	7	17	11	25	15	33	19	41	23	49
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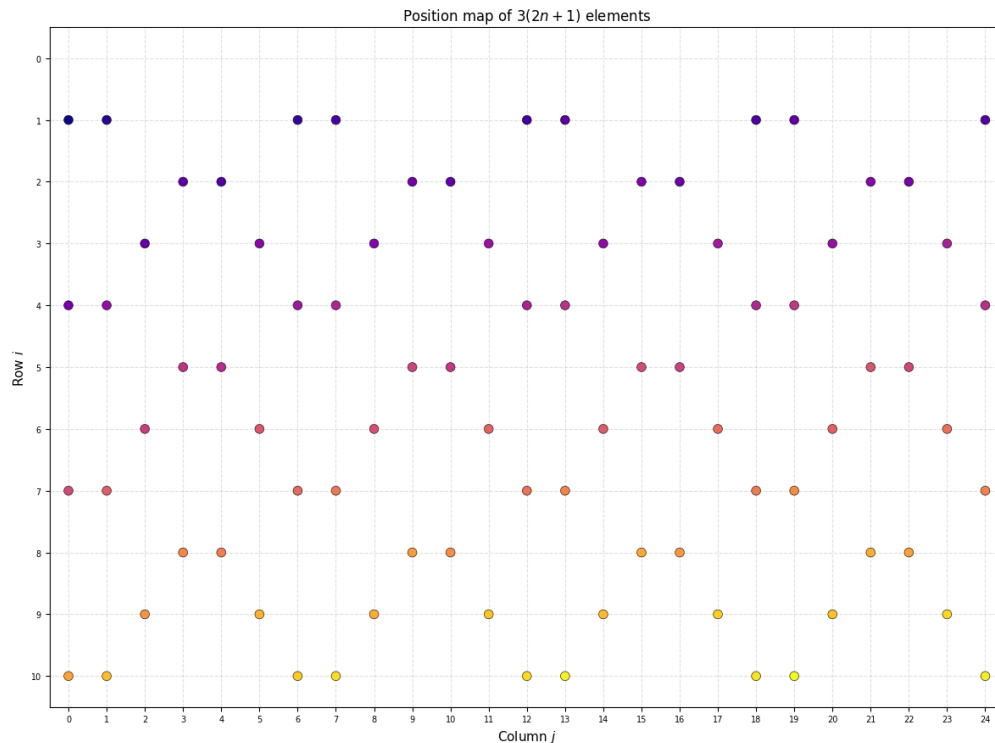
Order in Chaos



Question

Does the arithmetic structure of the sub-series $\{3(2n + 1)\}$ impose constraints on the global dynamics of the Collatz iteration, and does its transient behavior influence **convergence**?

Order in Chaos



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Scale it for large n

Parallel implementation

1

Matrix construction

2

Index Extraction

3

Maximum Identification



Scale it for large n

Parallel implementation

1

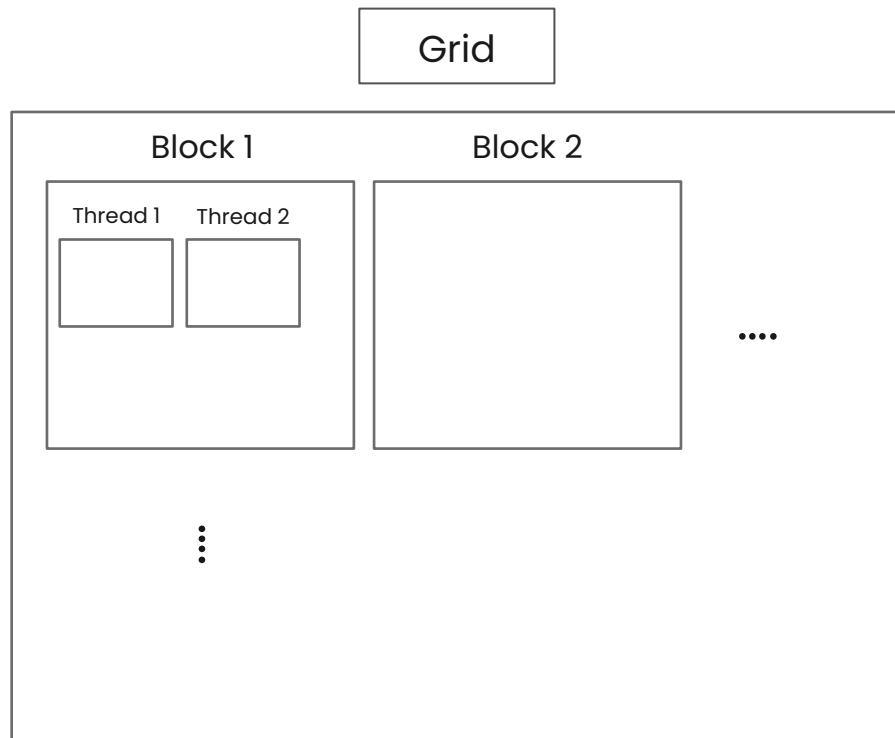
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3

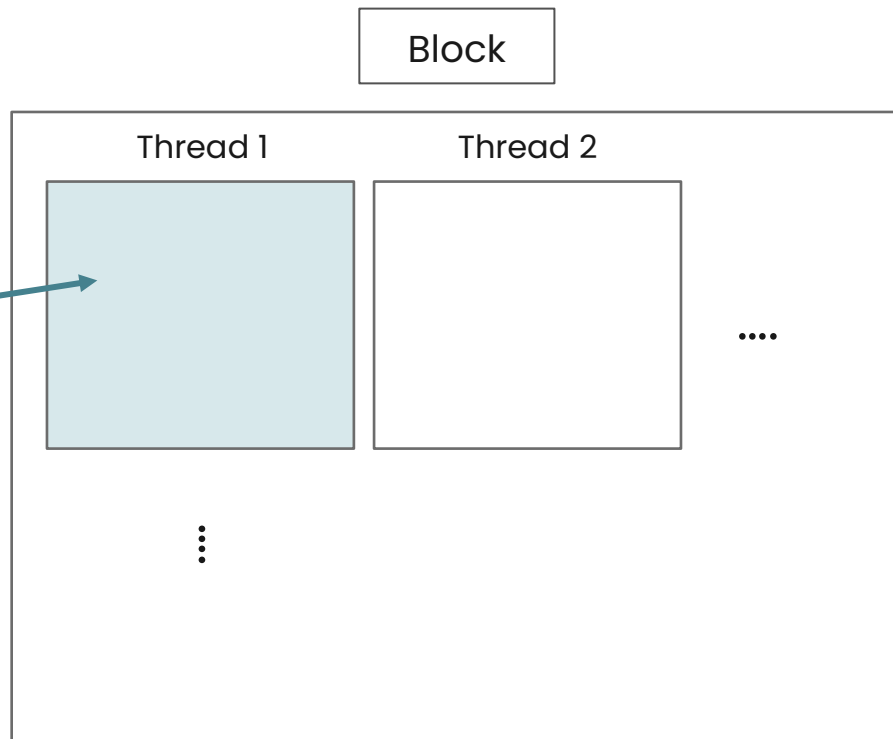
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Scale it for large n

Matrix construction

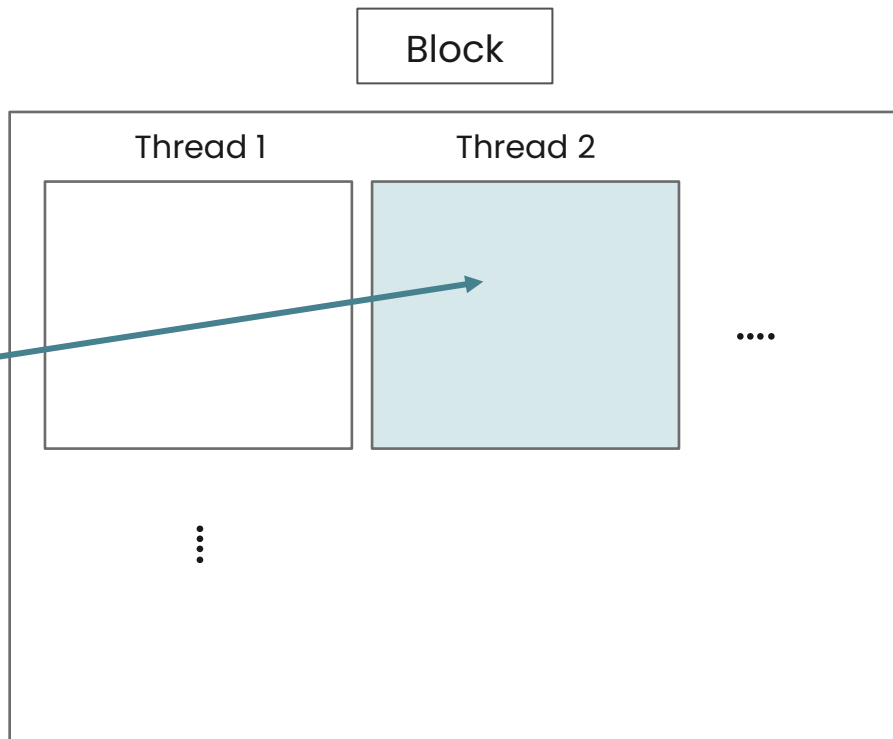
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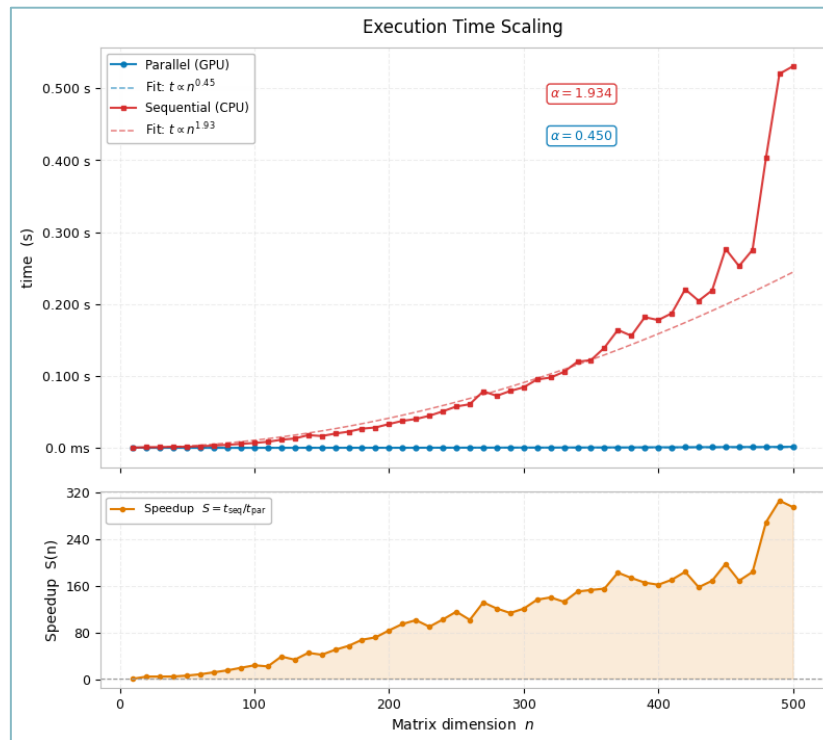


Results of the parallelism

the scalability behaviour of the parallel program as the input size grows

Table 5: Monitoring metrics as n increases ($n \times n$ matrix)

n	RAM (%)	Threads active	Threads launched	Max value
50	1.8	2,500	3,584	$\approx 2^{103.65}$
100	1.8	10,000	13,312	$\approx 2^{204.65}$
150	1.8	22,500	24,320	$\approx 2^{305.23}$
200	1.8	40,000	44,800	$\approx 2^{405.65}$
250	1.8	62,500	65,536	$\approx 2^{505.97}$
300	1.8	90,000	97,280	$\approx 2^{606.23}$
350	1.8	122,500	123,904	$\approx 2^{706.45}$
400	1.8	160,000	166,400	$\approx 2^{806.65}$
450	1.8	202,500	218,880	$\approx 2^{906.81}$
500	1.8	250,000	258,048	$\approx 2^{1006.97}$
550	1.8	302,500	317,952	$\approx 2^{1107.10}$

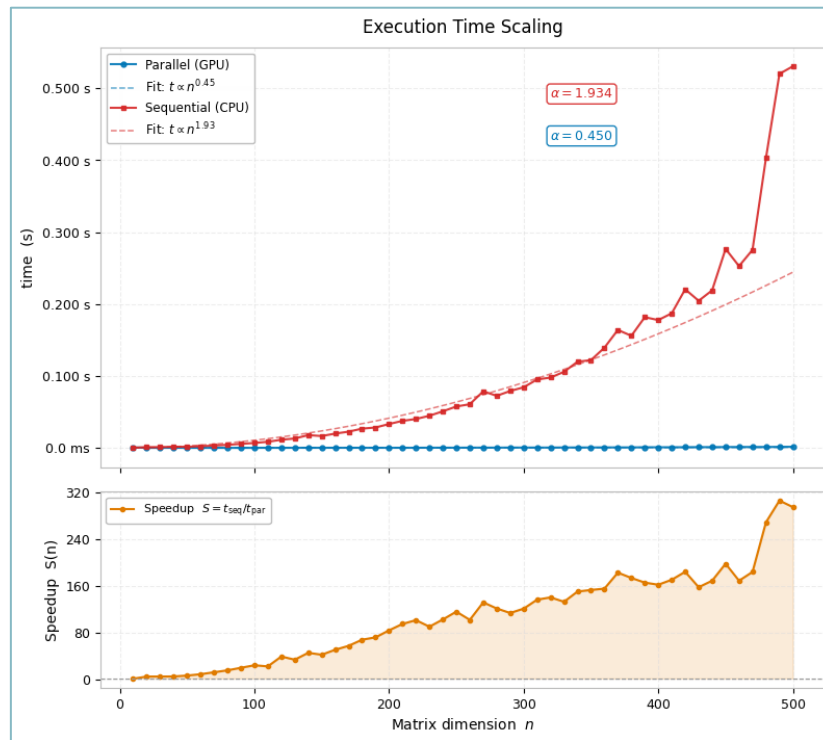


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Results of the parallelism



Max for 550 × 550 Matrix

134027868282182343102217216452227332869603767432395718358
902746451406220337057574336039547013608364641873131163880
656415844208536879822881651291114145142576531531168157458616
08495037132677768857225443818595307721222376801220890080
207172634937159753491435921170648913360737298984698093752
68396428704088067



Why are we still interested in Collatz conjecture

Cryptography applications

1

Collatz-inspired **chaos-based** hash function that achieves strong randomness and resistance to attacks.

Rasool & Belhaouari (2023). From Collatz conjecture to chaos and hash function. Chaos, Solitons & Fractals.

2

Another work proposes a symmetric cryptosystem based on Collatz sequences

Bandara & Ranasinghe (2022). A novel cryptographic scheme based on the Collatz conjecture. PIS Research Congress Proceedings.



THANKS!

Do you have any questions?

If you have a good GPU lying around
give it a try for bigger n and let us know the
maximum value you managed to reach

Github Repository: [HoceiniRihab/Collatz-Conjecture](#)

“Chaos is order yet undeciphered”

*José Saramago
The Double (Harvill, 2004)*